

Heterogeneous Nav2 placement: 50 m trials

Single A7 vehicle / single A15 per RSU; middle DVFS; device-wide overrun boost

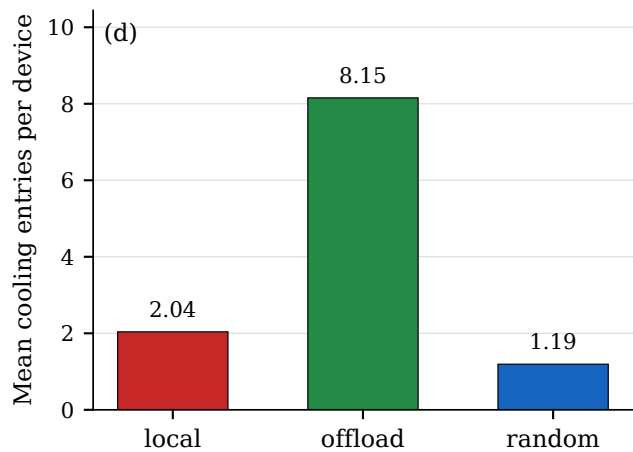
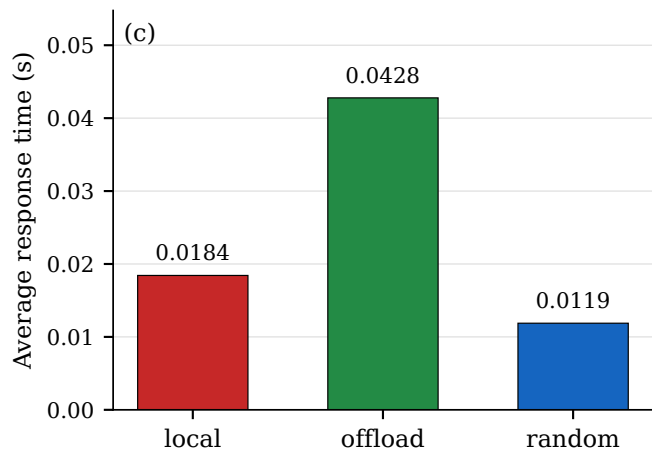
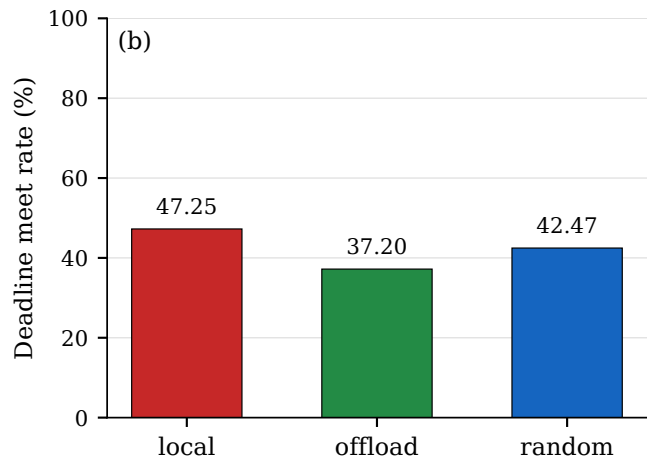
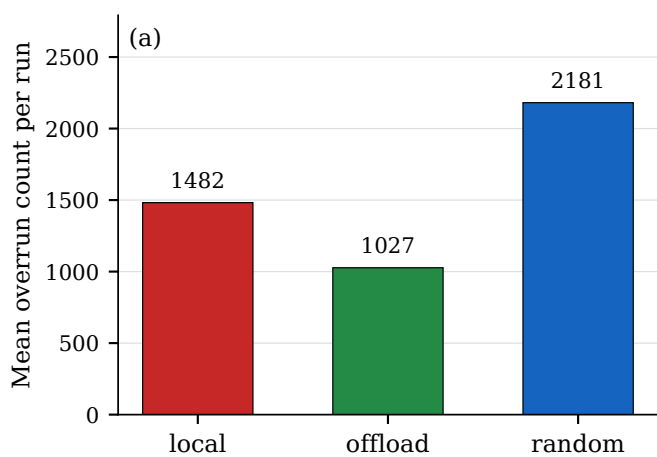
LOCAL (red): 50.024 m; 111.908 simulation s
18318 jobs; 18312 completed; 6 unfinished
Overruns: 1482; deadline meet rate: 47.248%
Mean completed-job response: 0.018428 s
Cooling entries: 53 total / 26 = 2.038462 per device
Validation errors: 0

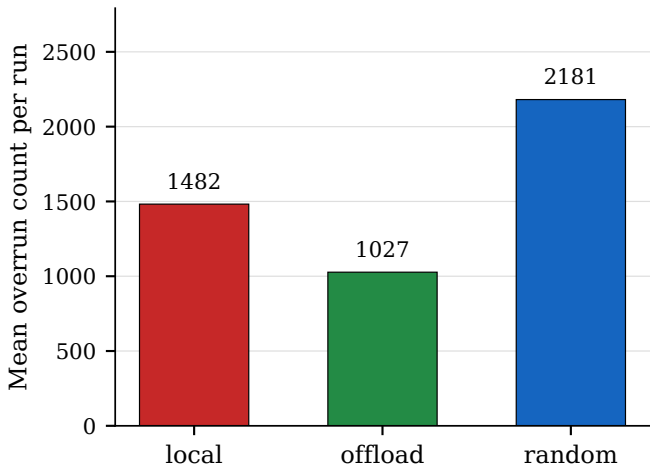
OFFLOAD (green): 50.014 m; 112.675 simulation s
11963 jobs; 11959 completed; 4 unfinished
Overruns: 1027; deadline meet rate: 37.198%
Mean completed-job response: 0.042780 s
Cooling entries: 212 total / 26 = 8.153846 per device
Validation errors: 0

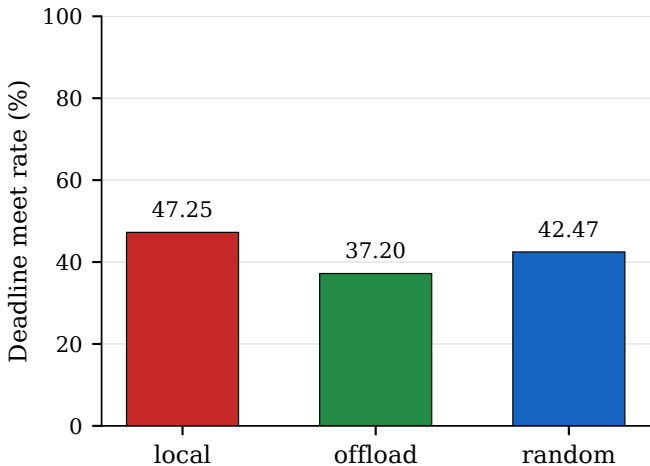
RANDOM (blue): 50.006 m; 111.797 simulation s
22350 jobs; 22349 completed; 1 unfinished
Overruns: 2181; deadline meet rate: 42.467%
Mean completed-job response: 0.011879 s
Cooling entries: 31 total / 26 = 1.192308 per device
Validation errors: 0

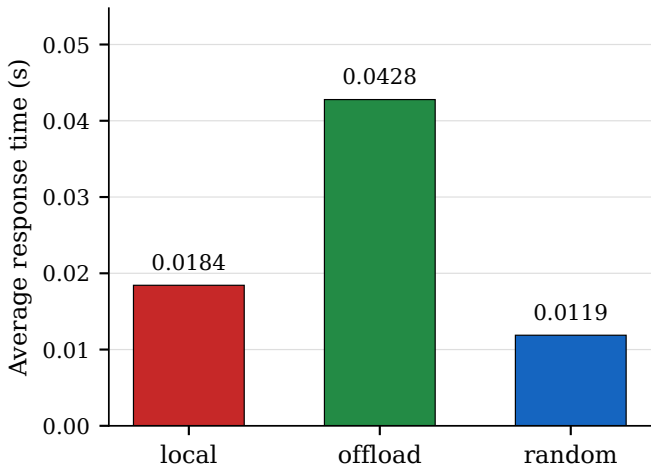
Pilot runs: [internal HOST-based replanning remains; see placement-diagnosis.en.md.](#)

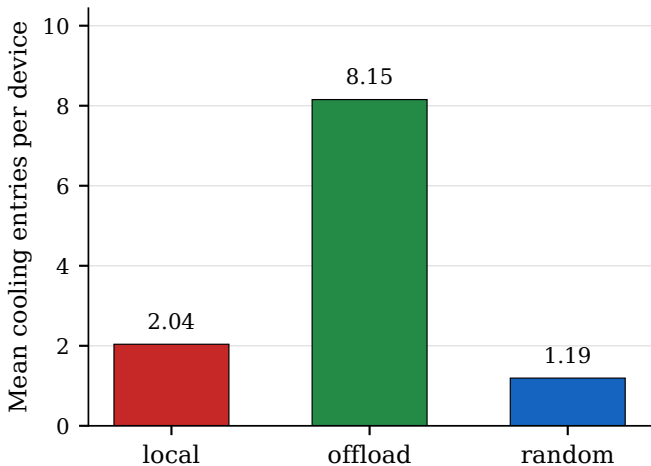
One run per policy: bars are descriptive; run-to-run variance is unavailable.
All deadlines and realized hardware values are shared across policies.
Real CPU measurements and actual callback arrivals can differ between runs.
Unfinished jobs are censored in mean response; due unfinished jobs count as misses.
The thermal metric counts whole-device cooling entries, not sampled hot ticks.

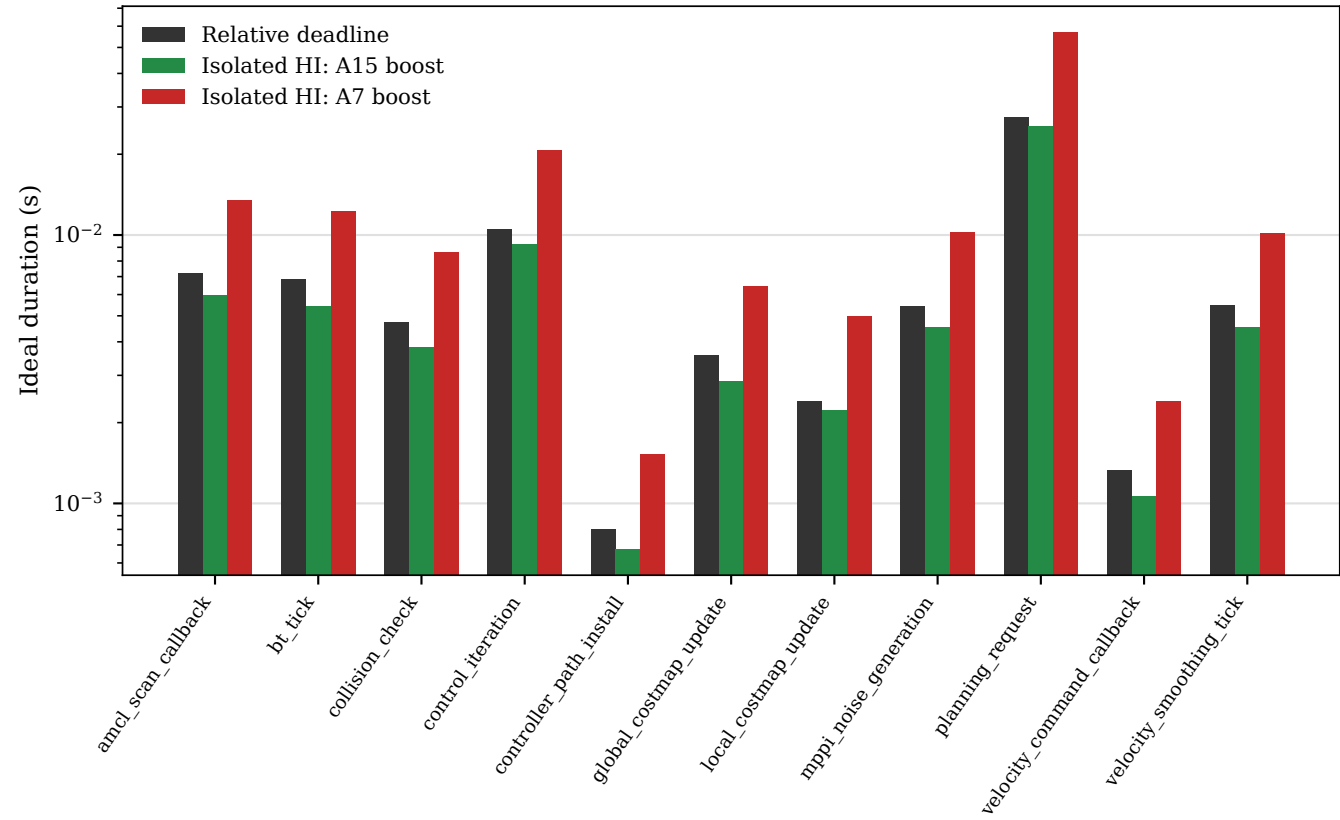












HI deadlines and processor-independent task work

$$W_i^{HI} = C_{i,ref}^{HI} f_{ref} \eta_{ref}, \quad D_i = U_i W_i^{HI} / (2000 \cdot 1.8), \quad d_j = r_j + D_i$$

Reference: 1000 MHz, eta = 1. U in [1.1, 1.3]. Periods remain seconds. No thermal costs in D.

Task	C ref LO (s)	C ref HI (s)	A15 max HI (s)	T (s)	D (s)
amcl scan callback	0.0048077	0.0199390	0.0055386	—	0.0071902
bt tick	0.0004929	0.0194456	0.0054016	0.010	0.0068513
collision check	0.0006137	0.0135562	0.0037656	—	0.0047424
control iteration	0.0076333	0.0306778	0.0085216	0.050	0.0105274
controller path install	0.0001770	0.0023685	0.0006579	—	0.0007997
global costmap update	0.0009884	0.0099746	0.0027707	1.000	0.0035696
local costmap update	0.0009596	0.0076756	0.0021321	0.200	0.0024004
mppi noise generation	0.0033925	0.0152440	0.0042344	—	0.0054141
planning request	0.0283547	0.0818079	0.0227244	—	0.0274828
velocity command callback	0.0000112	0.0038324	0.0010646	—	0.0013302
velocity smoothing tick	0.0001533	0.0162255	0.0045071	0.050	0.0054806

Execution: A7 1200 MHz / 1.0 V; A15 1500 MHz / 1.2 V. Normal execution uses the middle point; overrun boosts the whole device. The response calculation includes actual transfers, queueing and cooling; D excludes them. Completion uses 0.001 s ticks; D is unrounded. Full conversions are retained in task-timing-units.csv.